College of Computer at Al-Lith

Lecture notes on:

Discrete Structures I

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Course Title: Discrete Structures I

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Prerequisites : Introduction to math II

Level: Undergraduate (computer Science) / Year 2 (1^{st} Semester)

Semester: Fall 2019

Course Schedule:

Sunday : 3:00 PM-3:50 PM + 4:00 PM-4:50 PM
Tuesday: 3:00 PM-3:50 PM

Office Hours:

Sunday :10:00 AM-10:50 AM + 11:00 AM-11:50 AM

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Grading plan

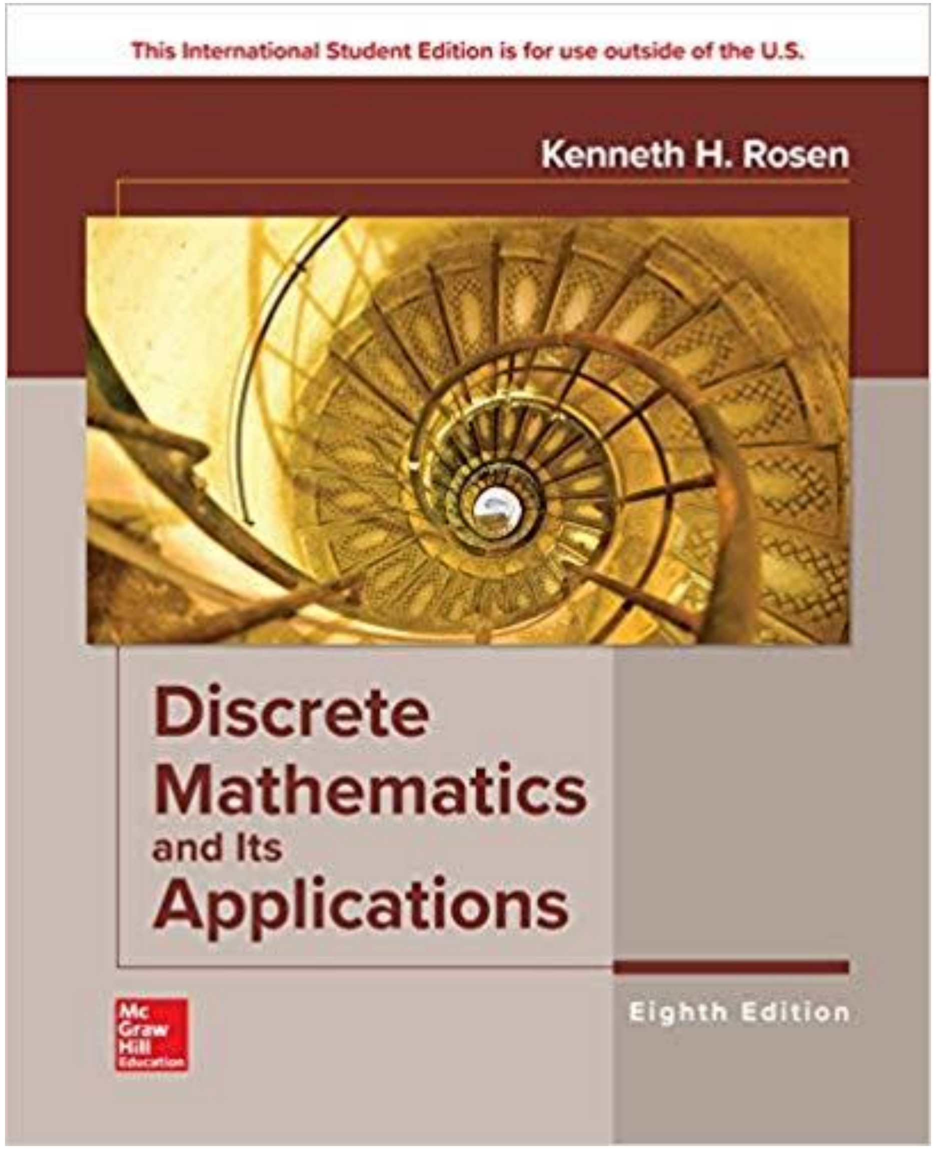
Midterm 1 : 25% (October 15th, 2019)
Midterm 2 : 25% (November 19th, 2019)
Final exam: 50% (December 2019 … TBA)

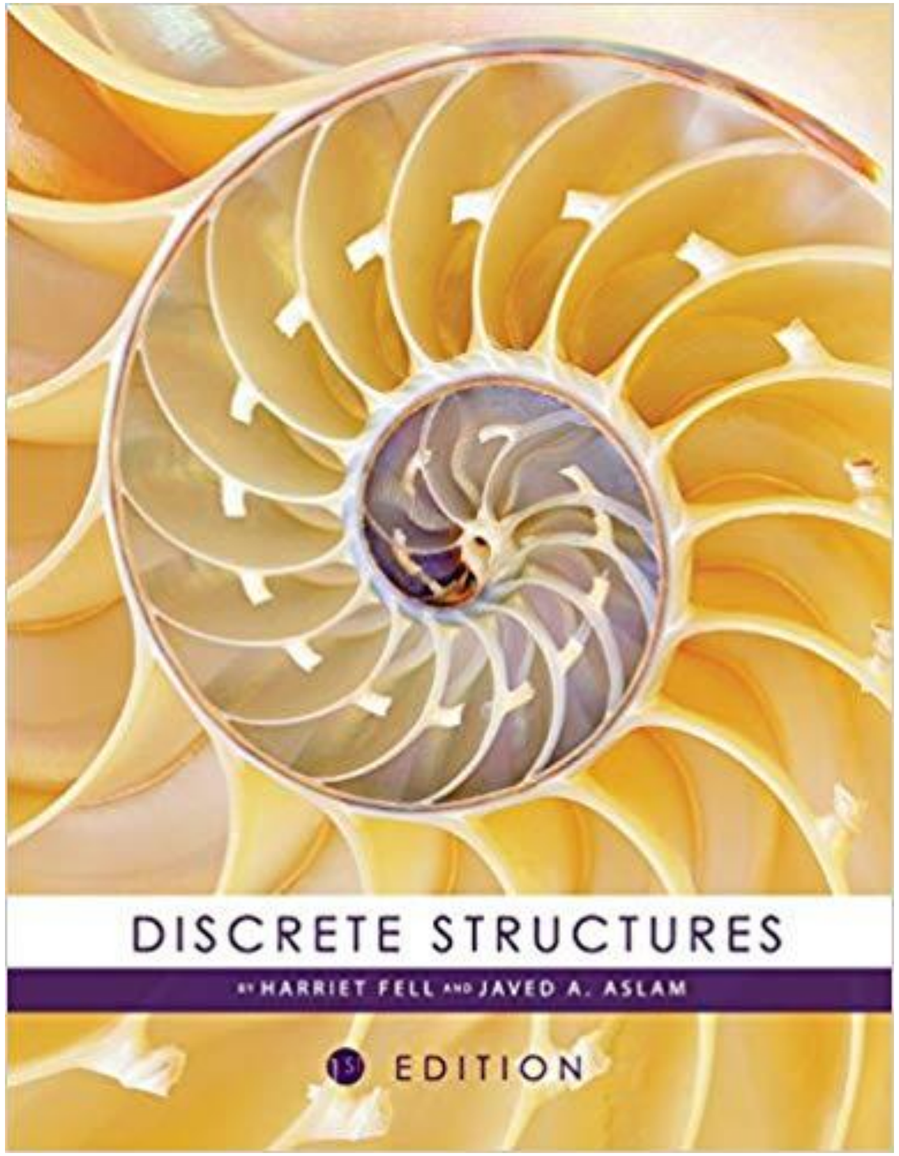
Textbook/References

- Discrete Mathematics and Its Applications, By Kenneth Rosen, Mc Graw Hill; Eighth Edition edition (2018)

- Discrete Structures, by Harriet Fell and Javed A. Aslam, Cognella Academic Publishing (2016)

Reference Books





CONTENTS

Chapter 1

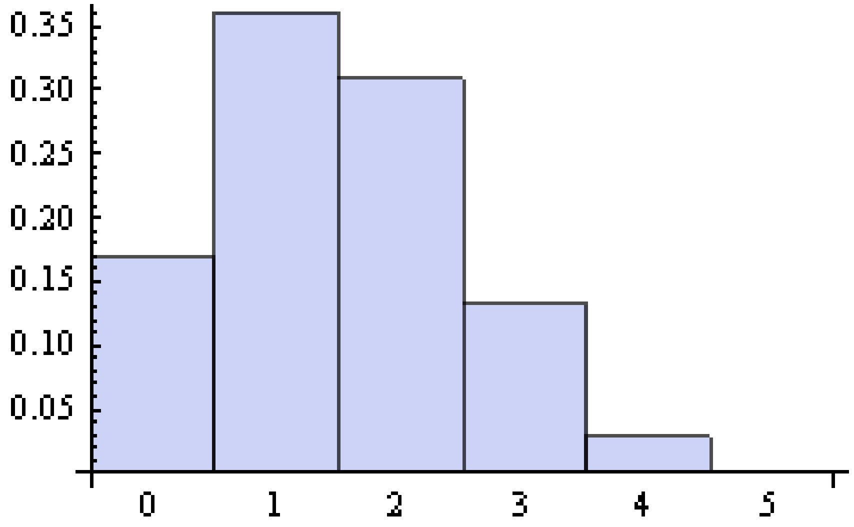
Introduction

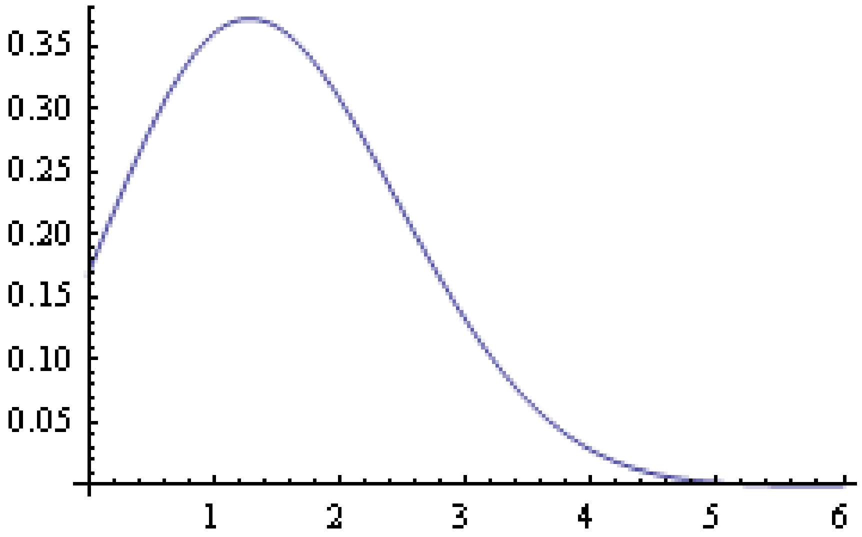
❑ What are Discrete Structures?

Structures that are naturally discrete and not continuous

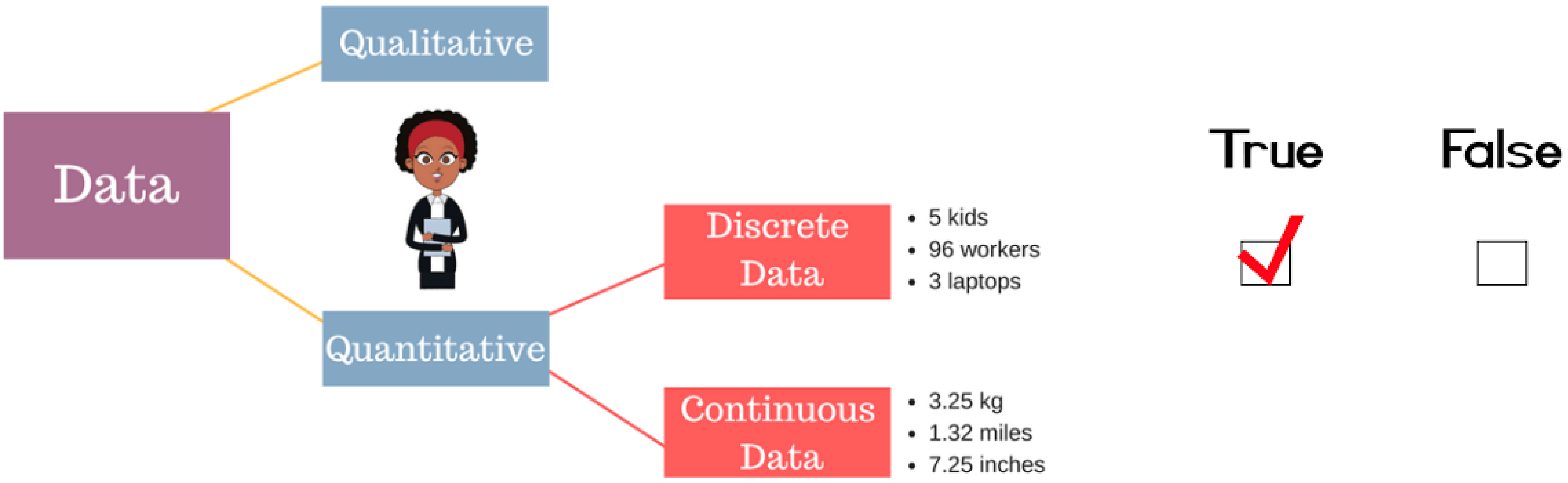
Integer are discrete. They have distinct values. They are non-continuous.

Real numbers are not discrete but continuous. They can vary smoothly and continuously.





❑ What are Discrete Structures?



We will learn the concepts associated with discrete objects, their properties, and relationships.

We can view it as the mathematics that is necessary for decision making in non-continuous situations

Why learn Discrete Structures?

- ✓The basis of all of digital information processing: 0,1
- ✓Analyze algorithms
- ✓Ability to realize how complex a problem/solution is
- ✓Software system design specification
- ✓Think "abstractly"
- ✓creating logical solution which is very useful in programming
- ✓Formal way of analyzing a problem
- ✓The mathematics of modern computer science is built almost entirely on discrete math
- ✓explore non-trivial “real world” problems that are challenging and interesting.
- ✓Discrete math teaches mathematical reasoning and proof techniques
- ✓Discrete math is fun

Use of Discrete Structures in Computer Science

- ✓ Networking
- ✓ Database
- ✓ Image Processing
- ✓ Programming Languages
- ✓ Compilers & Interpreters
- ✓ Software Engineering
- ✓ Artificial Intelligence
- ✓ Computer Architecture
- ✓ Operating Systems
- ✓ Security & Cryptography
- ✓ Advanced Algorithms & Data Structures
- ✓ Graphics & Animation
- ✓ ……

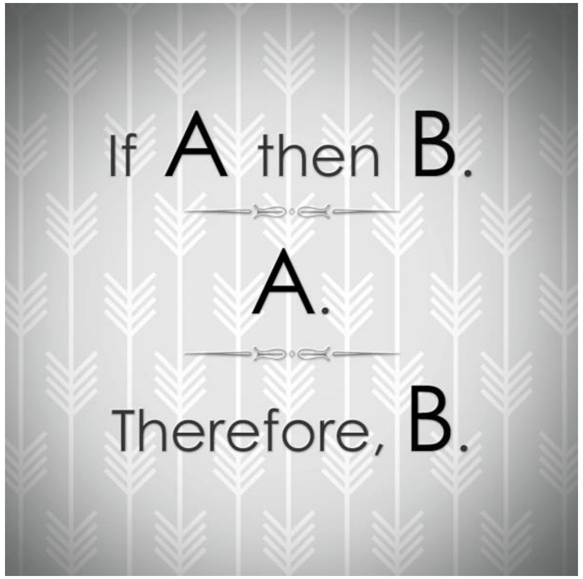
Chapter 2

Logic

What's Logic?

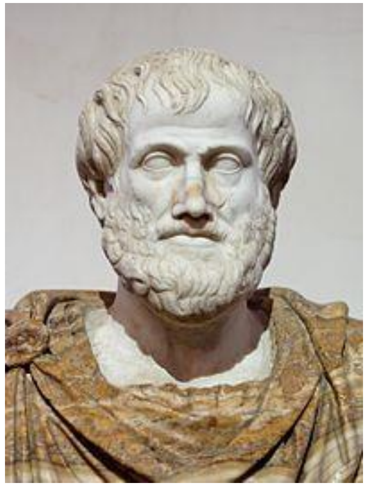
Logic is the study of the principles and methods that distinguish between a valid and invalid argument.

Methods of reasoning, provides rules and techniques to determine whether an argument is valid

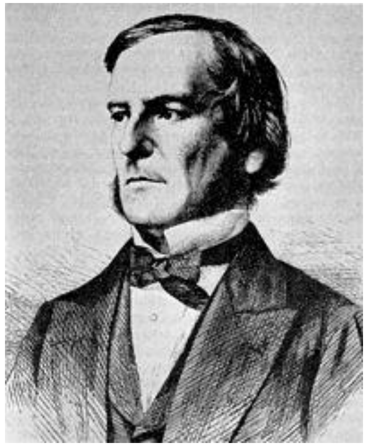


History …

- Aristotle developed propositional logic over 2000 years ago….



- George Boole wrote “The Mathematical Analysis of Logic” in 1848



Propositional Logic

- **Proposition**: *A proposition* is a declarative statement (a statement that declares a fact) that is either TRUE or FALSE, but not both.

- The area of logic that deals with propositions is called propositional logic.

Propositions	Not Propositions
1. Riyadh is the capital of Saudi Arabia	1. How many students in this class?
2. It's raining now.	2. Bring me coffee!
3. 1 + 1 = 2	3. X + 2 = 3
4. 4 + 3 = 10	4. Y + Z = X

Propositional Logic

- Propositions are denoted by letters i.e. p, q, r, s.

- Truth value of true proposition is denoted by T
- Truth value of false proposition is denoted by F

- An operator or connective combines one or more operand expressions into a larger expression. (E.g., "+" in numeric exprs.)
  - Unary operators take 1 operand (e.g., -3);
  - binary operators take 2 operands (eg 3×4).

- Propositional or Boolean operators operate on propositions (or their truth values) instead of on numbers.

Propositional Logic

Logic operators:

Formal Name	Nickname	Arity	Symbol
Negation operator	NOT	Unary	$\neg$
Conjunction operator	AND	Binary	$\wedge$
Disjunction operator	OR	Binary	$\vee$
Exclusive-OR operator	XOR	Binary	$\oplus$
Implication operator	IMPLIES	Binary	$\rightarrow$
Equivallence operator	IFF	Binary	$\leftrightarrow$

Propositional Logic

Negation: Suppose p is a proposition.

The negation of p is written $\neg$p and has meaning: “It is not the case that p.”

The proposition $\neg$p is read “NOT p”

Example: p: “Today is Friday”

$\neg$p: “Today is NOT Friday”

Truth table for negation:

p	$\neg p$
F	T
T	F

Propositional Logic

Conjunction corresponds to English "and."

Conjunction: Let p and q be two propositions. The conjunction of p and q, denoted by $p \wedge q$, is the proposition "p and q".

Example: p: "Today is Friday"

q: "It is raining today"

$p \wedge q$: "Today is Friday and it is raining today"

Truth table for conjunction:

p	q	$p \wedge q$
F	F	F
F	T	F
T	F	F
T	T	T

Propositional Logic

Disjunction corresponds to English "or."

Disjunction: Let p and q be two propositions. The disjunction of p and q, denoted by $p \vee q$, is the proposition "p or q".

Example: p: "Ali is tall"

q: "Sami is tall"

$p \vee q$: "Ali is tall or Sami is tall"

Truth table for disjunction:

Propositional Logic

Exclusive Or: Let p and q be two propositions.

The exclusive or of p and q, denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true, otherwise false.

Truth table for Exclusive Or:

p	q	$p \oplus q$
F	F	F
F	T	T
T	F	T
T	T	F

Propositional Logic

Implication: it corresponds to English “if p then q” or “p implies q.”

Let p and q be two propositions. The conditional statement (implication) $p \rightarrow q$ is the proposition “if p, then q”.

Example:

If it is raining, then it is cloudy.

If you get 100% in the final, then

you will get Grade A+.

Truth table for implication:

p	q	$p \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

Propositional Logic

Equivalence: If p and q are statement variables,

the biconditional of p and q is p $\leftrightarrow$ q states that p is true if and only if (IFF) q is true.

When we say P if and only if q , we are saying that p says the same thing as q.

$p \leftrightarrow q$ means that p and q have the **same** truth value.

Truth table for equivalence:

P	Q	P $\leftrightarrow$ Q
T	T	T
T	F	F
F	T	F
F	F	T

Propositional Logic

Compound Propositions

- A string consisting of variables, parenthesis and connectives is called a compound proposition.

- Example
- $\neg(p \lor q)$, $\neg(p \land q)$, $(p \land (p \rightarrow q \land r))$

Note: "2^n" is used to create a truth table for the given expression. Where n is the number of variable used in the expression.

Propositional Logic

Exercise: Draw the truth table of the following expressions

- $(P \vee Q) \wedge \neg R$
- $P \rightarrow (\neg P \rightarrow Q)$
- $(P \rightarrow Q) \leftrightarrow (\neg P \vee Q)$

Propositional Logic

Special definitions

Converse: $q \rightarrow p$ is converse of $p \rightarrow q$.

Ex.: $p \rightarrow q$: "If it is noon, then I am hungry."

$q \rightarrow p$: "If I am hungry, then it is noon."

Contrapositive: $\neg q \rightarrow \neg p$ is contrapositive of $p \rightarrow q$.

Ex.: $p \rightarrow q$: "If it is noon, then I am hungry."

$\neg q \rightarrow \neg p$: "If I am not hungry, then it is not noon."

Inverse: $\neg p \rightarrow \neg q$ is inverse of $p \rightarrow q$.

Ex.: $p \rightarrow q$: "If it is noon, then I am hungry."

$\neg p \rightarrow \neg q$: "If it is not noon, then I am not hungry."

Propositional Logic

Special definitions

A compound proposition that is:

(1) always true is called a **tautology**

(2) always false is called a **contradiction**

(3) neither a tautology nor a contradiction is called **contingency** or satisfiable.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
F	T	T	F
T	F	T	F

Propositional Logic

Exercise: which of the following expressions is a tautology, contradiction or contingency?

- ❑ $(P \rightarrow Q) \wedge (Q \rightarrow R)$
- ❑ $P \wedge Q \rightarrow P \vee Q$
- ❑ $P \wedge \neg P$

Propositional Logic

Exercise: Let

p="It rained last night",

q="I was at home last night,"

r="The grass was wet this morning."

Translate each of the following into English:

- $\neg p$ =
- $r \wedge \neg p$ =
- $\neg r \vee p \vee q$ =

Propositional Logic

Logical equivalence

The compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.

In other words, p and q are logically equivalent if their truth tables are the same.

We write $p \equiv q$.

Example: $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

Truth tables for $\neg(p \vee q)$ and $\neg p \wedge \neg q$:

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
F	F	F	T	T	T	T
T	F	T	F	F	T	F
F	T	T	F	T	F	F
T	T	T	F	F	F	F

Propositional Logic

Exercise: Show that the following expressions are equivalent

- $p \to q \equiv \neg p \vee q.$
- $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

Propositional Logic

Standard logical equivalences

- Identity Law: $p \wedge T \equiv p$ and $p \vee F \equiv p$
- Idempotent Law: $p \vee p \equiv p$ and $p \wedge p \equiv p$
- Domination Law: $p \vee T \equiv T$ and $p \wedge F \equiv F$
- Negation Law: $p \vee \neg p \equiv T$ and $p \wedge \neg p \equiv F$
- Double Negation Law: $\neg(\neg p) \equiv p$
- Commutative Law: $p \vee q \equiv q \vee p$ and $p \wedge q \equiv q \wedge p$
- Associative Law: $(p \vee q) \vee r \equiv p \vee (q \vee r)$ and $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- Distributive Law: $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ and $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- Absorption Law: $p \vee (p \wedge q) \equiv p$ and $p \wedge (p \vee q) \equiv p$
- De Morgan's Law: $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$
- Implication Equivalence: $p \to q \equiv \neg p \vee q$
- Biconditional Equivalence: $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

Propositional Logic

Logical equivalences involving Implications

$$p \to q \equiv \neg p \vee q$$

$$p \to q \equiv \neg q \to \neg p$$

$$p \vee q \equiv \neg p \to q$$

$$p \wedge q \equiv \neg (p \to \neg q)$$

$$\neg (p \to q) \equiv p \wedge \neg q$$

$$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$$

$$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$$

$$(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$$

$$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$$

Logical Equivalences involving Biconditionals

$$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg (p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Propositional Logic

Exercise: prove the following equivalences using equivalence rules

- ❑ $\neg (p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$
- ❑ $\neg (\neg p \wedge q) \wedge (p \vee q) \equiv p$

Propositional Logic

Rules of inference

- An **axiom** is a statement that is given to be true.
- A **rule of inference** is a logical rule that is used to deduce one statement from others.
- A **theorem** is a proposition that can be proved using definitions, axioms, other theorems, and rules of inference.
- **Formal Proofs**. To prove an argument is valid:

Assume the hypotheses are true. Use the rules of inference and logical equivalences to show that the conclusion is true.

$$[h1 \wedge h2 \wedge h3 \wedge \ldots \wedge hn] \rightarrow C$$

Propositional Logic

Rules of inference

inference rule	tautology	name
$\begin{array}{rl} & p \\ & p \to q \\ \hline \therefore & q \end{array}$	$(p \land (p \to q)) \to q$	Modus ponens (mode that affirms)
$\begin{array}{rl} & \neg q \\ & p \to q \\ \hline \therefore & \neg p \end{array}$	$(\neg q \land (p \to q)) \to \neg p$	Modus tollens (mode that denies)
$\begin{array}{rl} & p \to q \\ & q \to r \\ \hline \therefore & p \to r \end{array}$	$((p \to q) \land (q \to r)) \to (p \to r)$	hypothetical syllogism
$\begin{array}{rl} & p \lor q \\ & \neg p \\ \hline \therefore & q \end{array}$	$((p \lor q) \land (\neg p)) \to q$	disjunctive syllogism
$\begin{array}{rl} & p \\ \hline \therefore & p \lor q \end{array}$	$p \to (p \lor q)$	addition
$\begin{array}{rl} & p \land q \\ \hline \therefore & p \end{array}$	$(p \land q) \to p$	simplification
$\begin{array}{rl} & p \\ & q \\ \hline \therefore & p \land q \end{array}$	$((p) \land (q)) \to (p \land q)$	conjunction
$\begin{array}{rl} & p \lor q \\ & \neg p \lor r \\ \hline \therefore & q \lor r \end{array}$	$((p \lor q) \land (\neg p \lor r)) \to (q \lor r)$	resolution

Propositional Logic

Example

- **Assumptions:** $\neg p \wedge q,\ r \to p,\ \neg r \to s,\ s \to t$
- **We want to show:** t

Proof:

- 1. $\neg p \wedge q$ Hypothesis
- 2. $\neg p$ Simplification
- 3. $r \to p$ Hypothesis
- 4. $\neg r$ Modus tollens (step 2 and 3)
- 5. $\neg r \to s$ Hypothesis
- 6. s Modus ponens (steps 4 and 5)
- 7. $s \to t$ Hypothesis
- 8. t Modus ponens (steps 6 and 7)
- **end of proof**

Propositional Logic

Example

Prove the following:

- 1. If the dog eats the cat food or scratches at the door, then the parrot will bark.
- 2. If the cat eats the parrot, then the parrot will not bark.
- 3. If the cat does not eat the parrot, then it will eat the cat food.
- 4. The cat did not eat the cat food.
- 5. Therefore, the dog does not eat the cat food either.

Propositional Logic

1. Assign propositional variables to the component propositions in the argument:

- d = the dog eats the cat food
- s = the dog scratches at the door
- p = the parrot will bark
- c = the cat eats the parrot
- e = the cat eats the cat food

2. Represent the formal argument using the variables:

$$\begin{array}{l} (d \vee s) \rightarrow p \\ c \rightarrow \neg p \\ \neg c \rightarrow e \\ \neg e \\ \hline \therefore \neg d \end{array}$$

Propositional Logic

3. Use the hypotheses, the rules of inference, and any logical equivalences to prove that the argument is valid:

	Assertion	Reason
1.	$\neg c \to e$	hypothesis 3
2.	$\neg e$	hypothesis 4
3.	c	steps 1 and 2 and *modus tollens*
4.	$c \to \neg p$	hypothesis 2
5.	$\neg p$	steps 3 and 4 and *modus ponens*
6.	$(d \vee s) \to p$	hypothesis 1
7.	$\neg(d \vee s)$	steps 5 and 6 and *modus tollens*
8.	$\neg d \wedge \neg s$	step 7 and De Morgan's law
9.	$\neg d$	step 8 and conjunction elimination

Propositional Logic

Exercise: Give a formal proof that the following arguments are valid.

$A \vee B$

$\neg C \rightarrow \neg B$

$\neg A$

$\therefore \ C$

$\neg (A \vee B) \rightarrow C$

$\neg C$

$\neg A$

$\therefore \ B$

Propositional Logic

Predicate Logic

Some English statements are hard to model in propositional logic:

- *All students are present*
- *There exists a prime number greater than 10*
- *$x>1$*

The propositional logic is not powerful enough to represent all types of assertions that are used in computer science and mathematics,

Thus we need more powerful logic to deal with these and other problems. **The predicate logic** (First order logic –FOL) is one of such logic and it addresses these issues among others.

Predicate Logic

- A predicate is modeled as a function P(·) from objects to propositions.

  P(x) = "x is sleeping" (where x is any object).

  We can not decide whether P(x) is true or false

- The result of applying a predicate P to an object x=a is the proposition P(a).

  e.g. if P(x) = "x > 1",
  then P(3) is the proposition "3 is greater than 1."

- Note: The predicate P itself (e.g. P="is sleeping") is not a proposition (not a complete sentence).

Predicate Logic

- Predicate logic includes propositional functions of any number of arguments.

  e.g. let P(x,y,z) = “x gave y the grade z”,

  x=“Mike”, y=“Mary”, z=“A”,

  P(x,y,z) = “Mike gave Mary the grade A.”

- The collection of values that a variable x can take is called x’s universe of discourse.

  e.g., let P(x)=“x+1>x”.

  we could define the course of universe as the set of integers.

Predicate Logic

Quantifier

- $P(x)$
  - Which x makes P hold?
  - Does it hold for any x?
  - Does it only hold for some x?

- *Quantifier*
  - $\forall x P(x)$
    - *Universal quantifier*
    - For all x, $P(x)$ holds.
  - $\exists x P(x)$
    - *Existential quantifier*
    - For some x, $P(x)$ holds.
    - There exists x which makes $P(x)$ hold.

- $Q(x)$ = "x is mortal"
  - $\forall x Q(x)$ = "Everybody is mortal"
  - $\exists x Q(x)$ = "Someone is mortal", "There is someone who is mortal"

Predicate Logic

Syntax of predicate logic (FOL)

• Constants	KingJohn, 2, NUS,...
• Predicates	Brother, >,...
• Functions	Sqrt, LeftLegOf,...
• Variables	x, y, a, b,...
• Connectives	$\neg, \Rightarrow, \wedge, \vee, \Leftrightarrow$
• Equality	=
• **Quantifiers**	$\forall, \exists$

Predicate Logic

Suppose we have the predicate *E(x)* meaning *x* is even, O(x) meaning x is odd and I(x) meaning x is an integer

and that the universe is the set of integers

- "Not every integer is even" = $\neg \forall x \, \mathrm{E}(x)$
- "Some integers are not even" = $\exists x \neg \mathrm{E}(x)$
- "Some integers are even and some are odd" = $\exists x \; \mathrm{E}(x) \wedge \exists x \; \mathrm{O}(x)$
- "If an integer is not even, then it is odd" = $\forall x \; [\neg \mathrm{E}(x) \rightarrow \mathrm{O}(x)]$
- "All integers are even" = $\forall x \; [\mathrm{I}(x) \rightarrow \mathrm{E}(x)]$
- "Some integers are odd" = $\exists x \; [\mathrm{I}(x) \wedge \mathrm{E}(x)]$
- "Only integers are even" = $\forall x \; [\mathrm{E}(x) \rightarrow \mathrm{I}(x)]$

Predicate Logic

Exercise

Translate the following predicates to English language:

S(x) : x is a student

E(x) : x enjoy discrete structures

$\forall x \ [S(x) \rightarrow E(x)]$

S(x) : x is a student

C(x) : x has a car

$\exists x \ [S(x) \wedge C(x)]$

E(x,y) : x eats y

F(x) : x is a food

$\forall x \ \forall y \ E(x,y) \rightarrow F(y)$

Predicate Logic

Order of quantifier

The order of quantifiers is important; they may not commute.

$$(1)\ \forall x \forall y P(x,y) \Leftrightarrow \forall y \forall x P(x,y), \text{ and}$$
$$(2)\ \exists x \exists y P(x,y) \Leftrightarrow \exists y \exists x P(x,y),$$
$$\text{but}$$
$$(3)\ \forall x \exists y P(x,y) \not\Leftrightarrow \exists y \forall x P(x,y).$$

L(x,y) : x Loves y

$\forall x \exists y\ E(x,y)$: everybody loves somebody

$\exists y \forall x\ E(x,y)$: there is someone who is loved by everyone

Predicate Logic

Equivalence

$$\forall x(P(x) \wedge Q(x)) \quad \equiv \quad \forall x P(x) \wedge \forall x Q(x)$$

$$\exists x(P(x) \vee Q(x)) \quad \equiv \quad \exists x P(x) \vee \exists x Q(x)$$

$$\forall x(P(x) \vee Q(x)) \quad \not\equiv \quad \forall x P(x) \vee \forall x Q(x)$$

$$\exists x(P(x) \wedge Q(x)) \quad \not\equiv \quad \exists x P(x) \wedge \exists x Q(x)$$

Predicate Logic

- Negation of existential quantifier

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

  - Negation is T if for every x, $P(x)$ is F.
  - Negation is F if there exists an x for which $P(x)$ is T.
- Negation of universal quantifier

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

  - Negation is T if there is an x for which $P(x)$ is F.
  - Negation is F if $P(x)$ is T for every x.

Predicate Logic

De Morgan's Laws for Quantifiers

- $\neg \forall x P(x) \Leftrightarrow \exists x \neg P(x)$
- $\neg \exists x P(x) \Leftrightarrow \forall x \neg P(x)$

Predicate Logic

Proofs

$$\neg\forall x(P(x) \rightarrow Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x)) \,.$$

$$
\begin{aligned}
& & & \neg\forall x(P(x) \rightarrow Q(x)) \\
\text{De Morgan} & & \equiv \quad & \exists x(\neg(P(x) \rightarrow Q(x))) \\
\text{impl-or} & & \equiv \quad & \exists x(\neg(\neg P(x) \vee Q(x))) \\
\text{De Morgan} & & \equiv \quad & \exists x(\neg\neg P(x) \wedge \neg Q(x)) \\
\text{double negation} & & \equiv \quad & \exists x(P(x) \wedge \neg Q(x)) \quad \square
\end{aligned}
$$

Thank you

End of Chapter 2

Chapter 3

Sets

Definitions

- A set is an unordered collection of objects.
- These objects are called the elements of the set.
- If a belongs to the set A, then we write $a \in A$, read “a is an element of A.”
- If a doesn’t belong to the set A, we write $a \notin A$, read “a is not an element of A.”
- Repeated elements in a set are ignored.
- The number of elements in a set A, also known as the cardinality of A, will be denoted by card (A) or $|A|$.
- If the set A has infinitely many elements, we write $|A| = \infty$.

Examples

- Let $D = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ be the set of the ten decimal digits.
- Then $4 \in D$ but $11 \notin D$.
- Also, $|D| = 10$.

- The sets $\{1, 2, 3\}$, $\{3, 2, 1\}$, and $\{1, 1, 1, 2, 2, 3\}$ actually represent the same set since repeated values are ignored and the order elements are listed does not matter.
- The cardinality of each of these sets is 3.

- **Exercise:** What is the set of prime numbers less than 10? Cardinality?

Equal sets

- We say two sets are equal if they contain the same elements.
- That is $\forall x(x \in A \leftrightarrow x \in B)$. If A and B are equal sets, we write $A = B$.

- Exercise: Let $A = \{1, 2, 3, 4, 5, 6\}$,
  $B = \{1, 2, 3, 4, 5, 4, 3, 2, 1\}$, and
  $C = \{6, 3, 4, 5, 1, 3, 2\}$.

Then

$|A| =?$,

$|B| =?$, and

$|C| =?$.

Which of A, B, and C represent the same sets?

Standard sets

The following notation is pretty standard, and we will follow it in this course.

- $\emptyset = \{\ \}$, the empty set.
- $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$, the non-negative integers
- $\mathbb{N}^+ = \{1, 2, 3, \ldots\}$, the positive integers
- $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2 \ldots\}$, the integers
- $\mathbb{Q} = \{q \mid q = a/b,\ a, b \in \mathbb{Z},\ b \neq 0\}$, the rational numbers
- $\mathbb{Q}^+ = \{q \mid q \in \mathbb{Q},\ q > 0\}$, the positive rationals
- $\mathbb{R}$, the real numbers

What's $\mathbb{C}$

Subsets

- If every element in A is also in B, we say that A is a subset of B and we write this as $A \subseteq B$.
- If $A \subseteq B$ and there is some $x \in B$ such that $x \notin A$, then we say A is a proper subset of B, denoting it by $A \subset B$.
- If there is some $x \in A$ such that $x \notin B$, then A is not a subset of B, which we write as $A \not\subseteq B$.
- **Example:** Let $S = \{1, 2, . . . , 20\}$, that is, the set of integers between 1 and 20, inclusive.

- Let $E = \{2, 4, 6, . . . , 20\}$, the set of all even integers between 2 and 20, inclusive. Notice that $E \subseteq S$.
- Let $P = \{2, 3, 5, 7, 11, 13, 17, 19\}$, the set of primes less than 20. Then $P \subseteq S$.

Subsets

Exercise: Let $S = \{n2 | n \in Z\}$ and $A = \{1, 4, 9, 16\}$. Answer each of the following,

- (a) Is $A \subseteq S$?
- (b) Is $A \subset S$?
- (c) Is $S \subseteq S$?
- (d) Is $S \subset S$?
- (e) Is $S \subset A$?

Exercise: Find all the subsets of $\{a, b, c\}$.

You can use either $\emptyset$ or $\{\}$ to denote the empty set, but not $\{\emptyset\}$.

The power set of a set

- The power set of a set is the set of all subsets of a set. The power set of a set A is denoted by P(A).

- Example: If A = {a, b, c}, example 5.20 implies that P(A) = {∅, {a}, {b}, {c}, {a, b}, {b, c}, {a, c}, {a, b, c}}.
- Notice that the solution is a set, the elements of which are also sets.
- An incorrect answer would be {∅, a, b, c, {a, b}, {b, c}, {a, c}, {a, b, c}}. This is incorrect
- because a is not the same thing as {a} (the set containing a). {a} ∈ P(A), but a $\notin$ P(A). This is a subtle but important distinction.

- Exercise: Find P({a, b, c, d}).

The power set of a set

Theorem: Let A be a set with n elements. Then $|P(A)| = 2n$.

Exercise: Let A be a set with 4 elements.

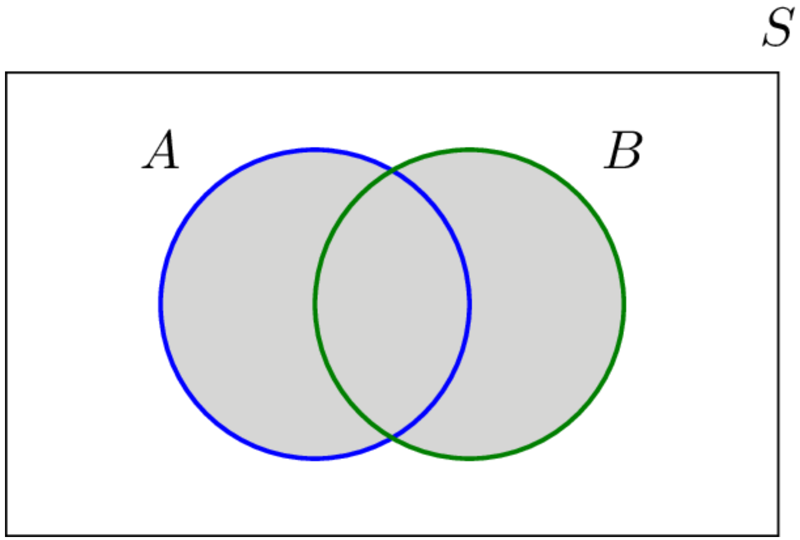
- (a) $|P(A)| =$
- (b) $|P(P(A))| =$
- (c) $|P(P(P(A)))| =$

Set Operations

- The union of two sets A and B is the set containing elements from either A or B.

- More formally, $A \cup B = \{x : x \in A \text{ or } x \in B\}$.

Notice that in this case the or is an inclusive or. That is, x can be in A, or it can be in B, or it can be in both.



Set Operations

- **Example:** Let A = {1, 2, 3, 4, 5, 6}, and B = {1, 3, 5, 7}.

Then AUB = {1, 2, 3, 4, 5, 6, 7}.

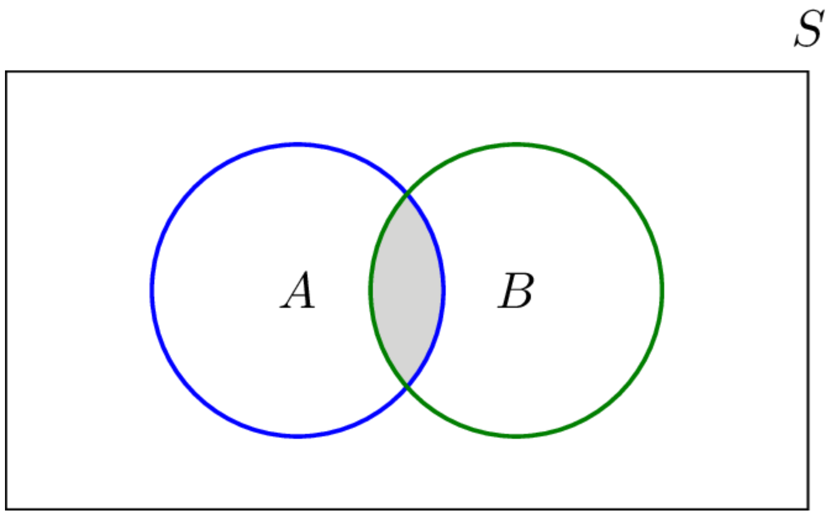
- **Exercise:** Let A be the set of even integers and B be the set of odd integers. Then

$A \cup B$=?

Set Operations

- The intersection of two sets A and B is the set containing elements that are in both A and B.

More formally, $A \cap B = \{x : x \in A \text{ and } x \in B\}$.



- Example: Let A = {1, 2, 3, 4, 5, 6}, and B = {1, 3, 5, 7, 9}. Then $A \cap B = \{1, 3, 5\}$.

- Exercise: Let A be the set of even integers and B be the set of odd integers. Then

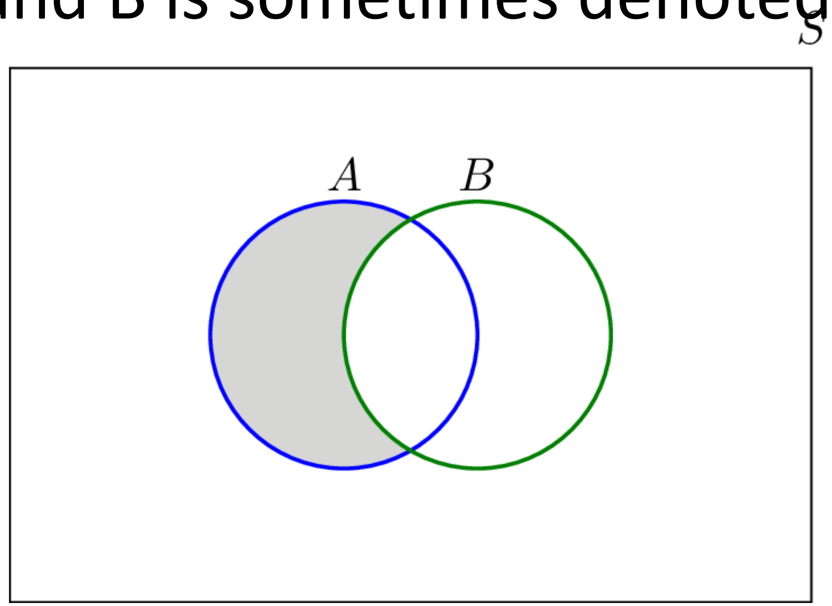
  $A \cap B = ?$

Set Operations

❑The difference (or set-difference) of sets A and B is the set containing elements from A that are not in B.

- More formally, $A \setminus B = \{x : x \in A \text{ and } x \; 6{\in} \; B\}$.

- The set difference of A and B is sometimes denoted by A − B.



Set Operations

Example: Let $A = \{1, 2, 3, 4, 5, 6\}$, and $B = \{1, 3, 5, 7, 9\}$.

Then $A \setminus B = \{2, 4, 6\}$ and $B \setminus A = \{7, 9\}$.

Exercise: Let A be the set of even integers and B be the set of odd integers.

Then

- $A \setminus B = ?$

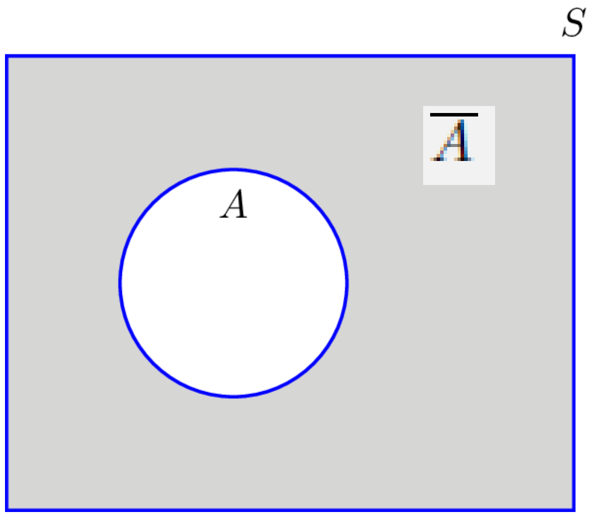
- $B \setminus A = ?$

Set Operations

❑Let A ⊆ U. The complement of A with respect to U is just the set difference U \ A.

More formally, $\overline{A}$ = {x ∈ U : x ∉ A} = U \ A.

In words, A is the set of everything not in A. Other common notations for set complement include A^c and A′.



Set Operations

- Example: Let U = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9} be the universal set of decimal digits and A = {0, 2, 4, 6, 8} ⊂ U be the set of even digits.

Then $\overline{A}$ = {1, 3, 5, 7, 9} is the set of odd digits.

- Exercise: Let A be the set of even integers and B be the set of odd integers, and let the universal set be U = Z.

Then

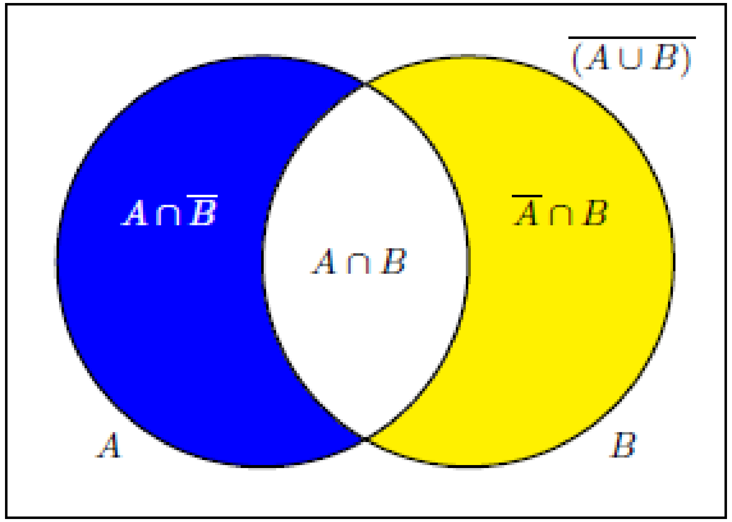
$\overline{A} =$

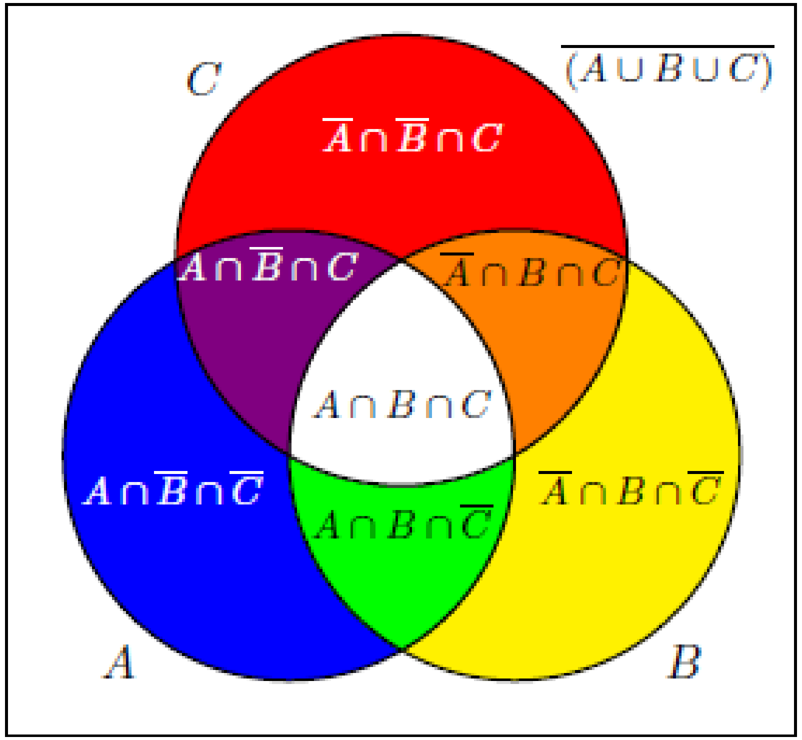
$\overline{B} =$

Set Operations

❑Theorem: Let A be a subset of some universal set U. Then

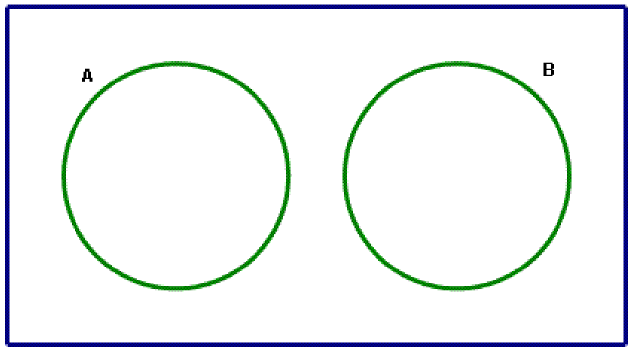
- $\overline{A} \cap A = \emptyset$, and
- $\overline{A} \cup A = U$.





Set Operations

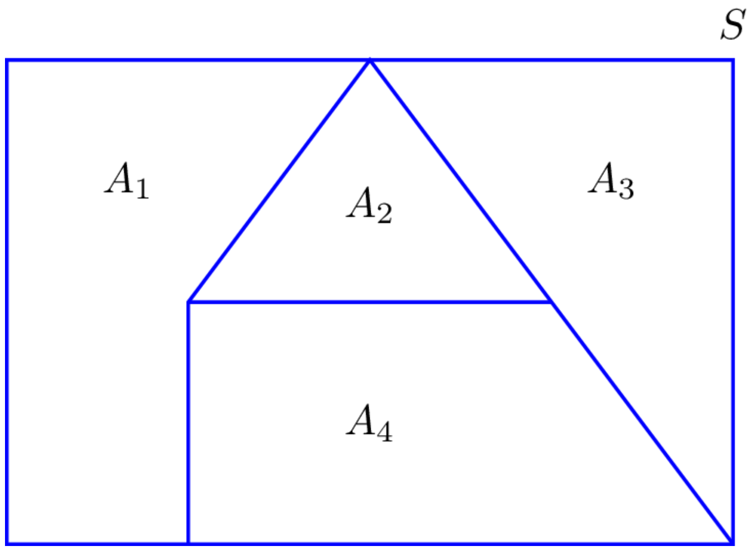
❑Two sets A and B are disjoint or mutually exclusive if $A \cap B$



That is, they have no elements in common.

Exercise: Let A be the set of even integers and B be the set of odd integers. Are A and B disjoint?

a collection of nonempty sets **A1,A2,⋯A1,A2,⋯** is a Partition of a set A if they are disjoint and their union is A.



Set identities

- Set identities can be used to show that two sets are the same.

Name	*Identity*
commutativity	$A \cup B = B \cup A$ $A \cap B = B \cap A$
associativity	$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$
distributive	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
identity	$A \cup \varnothing = A$ $A \cap U = A$
complement	$A \cup \overline{A} = U$ $A \cap \overline{A} = \varnothing$
domination	$A \cup U = U$ $A \cap \varnothing = \varnothing$
idempotent	$A \cup A = A$ $A \cap A = A$
complementation	$\overline{(\overline{A})} = A$
DeMorgan's	$\overline{A \cup B} = \overline{A} \cap \overline{B}$ $\overline{A \cap B} = \overline{A} \cup \overline{B}$
absorption	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$

Set identities

Theorem: Two sets A and B are equal if and only if $A \subseteq B$ and $B \subseteq A$.

Example: Prove that $A \setminus B = A \cap B$.

Proof:

……

……

Exercise: Of 40 people, 28 smoke and 16 chew tobacco. It is also known that 10 both smoke and chew. How many among the 40 neither smoke nor chew?

Set Operations

- The Cartesian product of sets A and B is the set $A \times B = \{(a, b) | a \in A \wedge b \in B\}$. In other words, it is the set of all ordered pairs of elements from A and B

Example: If $A = \{1, 2, 3\}$ and $B = \{a, b\}$, then

- $A \times B = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$, and
- $B \times A = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$.

Notice that $A \times B \neq B \times A$. If $A \neq B$, this is always the case.

Exercise: Let $A = \{1, 2, 3, 4\}$, and $B = \{3\}$. Compute $A \times B$.

$A \times B =$

Set Operations

❑Definition: If A is a set, then $A^2 = A \times A$, and $A^n = A \times A^{n-1}$.

Example: If $B = \{a, b\}$then

- $B^2 = \{(a, a), (a, b), (b, a), (b, b)\}$, and
- $B^3 = \{(a, a, a), (a, b, a), (b, a, a), (b, b, a), (a, a, b), (a, b, b), (b, a, b), (b, b, b)\}$

Exercise: Let $A = \{0, 1\}$. Find A^2 and A^3.

- $A^2 =$
- $A^3 =$

Thank you

End of Chapter 3

Chapter 4

Basic number theory

Number thoery

"Mathematics is the queen of sciences and number theory is the queen of mathematics." ... Carl Friedrich Gauss

Branch of mathematics concerned with properties of the positive integers (1, 2, 3, ...).

Several applications in computer science (security: hashing, cryptography, information coding, ...)

Topics:

- Divisibility
- Congruence
- Prime numbers

Divisibility

- Given integers *a* and *b* with **$b \neq 0$**, we say that *b* is a divisor or a factor of *a* and that *a* is divisible by *b* if and only if $a = q \times b$ for some integer q.

- We write ***b | a*** to signify that *a* is divisible by *b* and say "*b* divides *a*."

  - 3 is a divisor of 18
  - -7 is a divisor of 35

Divisibility

Theorem. for all $a, b, c \in \mathbf{Z}$:

1. $a|0$
2. $(a|b \wedge a|c) \rightarrow a \mid (b + c)$
3. $a|b \rightarrow a|bc$ for all integers c
4. $(a|b \wedge b|c) \rightarrow a|c$

Show that

- $3 \mid 126$
- $2 \mid 24$

Divisibility

Divisibility Rules	
A number is divisible by	
2	If last digit is 0, 2, 4, 6, or 8
3	If the sum of the digits is divisible by 3
4	If the last two digits is divisible by 4
5	If the last digit is 0 or 5
6	If the number is divisible by 2 and 3
7	cross off last digit, double it and subtract. Repeat if you want. If new number is divisible by 7, the original number is divisible by 7
8	If last 3 digits is divisible by 8
9	If the sum of the digits is divisible by 9
10	If the last digit is 0
11	Subtract the last digit from the number formed by the remaining digits. If new number is divisible by 11, the original number is divisible by 11
12	If the number is divisible by 3 and 4

Divisibility

Division Algorithm Theorem: Let *a* be an integer, and *d* be a positive integer. There are unique integers *q, r* with $r \in \{0,1,2,\ldots,d\text{-}1\}$ (ie, $0 \leq r < d$) satisfying

$$a = d \times q + r$$

- d is the divisor
- q is the quotient, q = a **div** d
- r is the remainder, r = a **mod** d

Example: 23 = 2 x 10 + 3

a = 23

d = 2

q =10

r = 3

The Greatest Common Divisor (gcd)

- Let a and b be integers not both of which are 0.

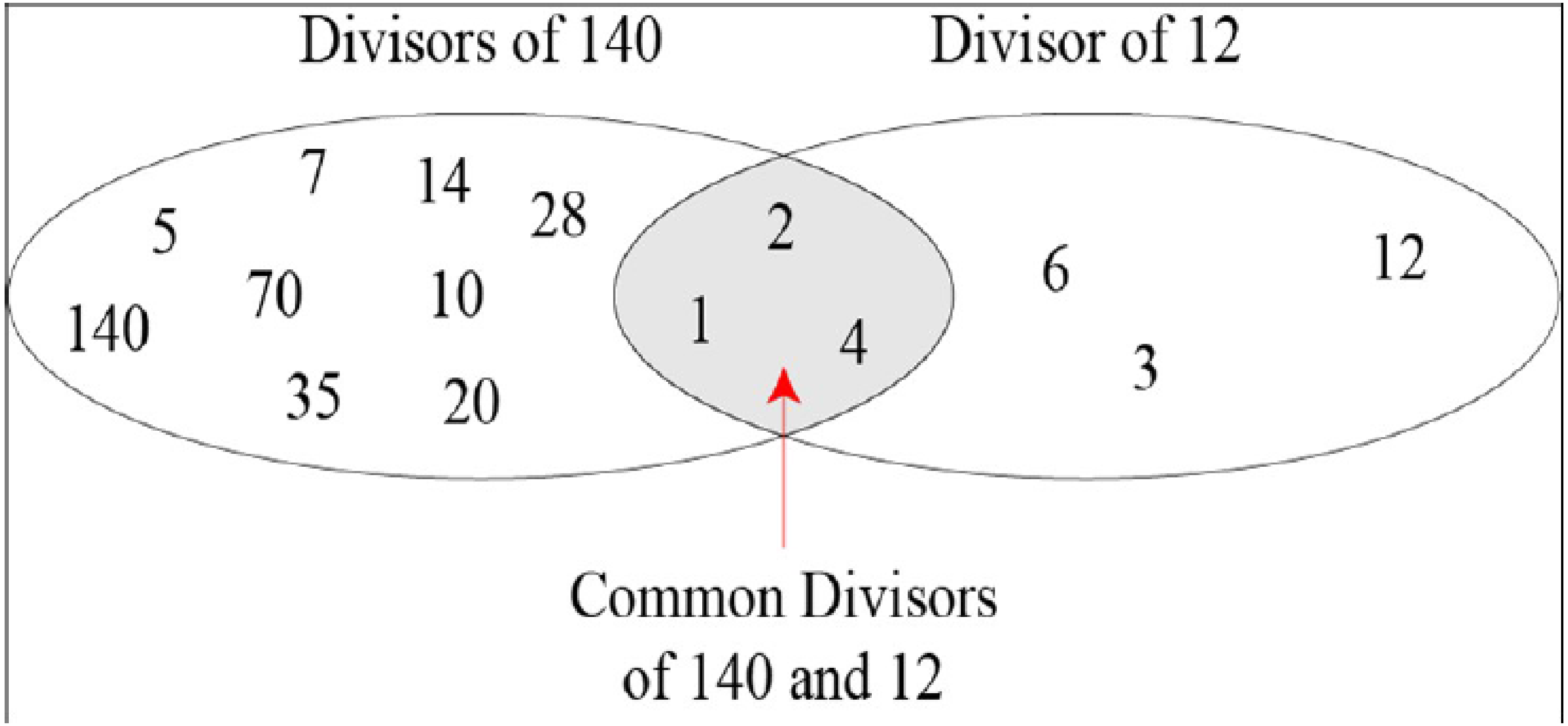
- An integer g is the **gcd** of a and b if and only if g is the largest common divisor of a and b; that is, if and only if

1. $g \mid a$, $g \mid b$ and
2. If c is any integer such that $c \mid a$ and $c \mid b$, then $c \leq g$.

Exercise: determine the gcd of 15 and 6.

Divisibility

Example: gcd(140,12)



The common divisors of 12 and 18 are$\{\pm1,\pm2,\pm3,\pm6\}$, and $\gcd(12, 18) = 6$.

Euclidean Algorithm

- Let a and b be natural numbers with b < a. To find the gcd of a and b, write

$$a = q_1 b + 1 \text{ with } 0 \leq r_1 < b$$

If $r_1 \neq 0$ write $b = q_2 r_1 + r_2$, with $0 \leq r_2 < r_1$

If $r_2 \neq 0$ write $r_1 = q_3 r_2 + r_3$, with $0 \leq r_3 < r_2$

If $r_3 \neq 0$ write $r_2 = q_4 r_3 + r_4$, with $0 \leq r_4 < r_3$

Continue the process until some remainder $r_{k+1} = 0$. Then the gcd of a and b is r_k, the **last nonzero remainder**.

Divisibility

Example: Show that gcd(4864, 3458) = 38 using Euclidian algorithm

$$
\begin{aligned}
4864 &= 1 \cdot 3458 + 1406 \\
3458 &= 2 \cdot 1406 + 646 \\
1406 &= 2 \cdot 646 + 114 \\
646 &= 5 \cdot 114 + 76 \\
114 &= 1 \cdot 76 + 38 \\
76 &= 2 \cdot 38 + 0.
\end{aligned}
$$

STOP

Exercise: gcd(299, 221) = ??? & gcd(287,91)=??? using Euclidian algorithm

Prime numbers

A positive integer $n > 1$ is called ***prime*** if it is only divisible by 1 and itself (i.e., only has 1 and itself as its positive factors).

Example: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 97

A number $n \geq 2$ which isn't prime is called ***composite***. (Iff there exists an a such that $a|n$ and $1 < a < n$)

Example:

All even numbers > 2 are composite.

By convention, 1 is neither prime or composite.

Primes

Fundamental Theorem of Arithmetic

Every positive integer greater than 1 has a *unique* representation as the product of a non-decreasing series of one or more primes

Examples:

- $2 = 2$
- $4 = 2 \cdot 2$
- $100 = 2 \cdot 2 \cdot 5 \cdot 5$
- $200 = 2 \cdot 2 \cdot 2 \cdot 5 \cdot 5$
- $999 = 3 \cdot 3 \cdot 3 \cdot 37$

Divisibility

- How do you check whether a positive integer n is prime?

  - Solution:

    Start testing to see if prime p divides n ($2|n$, $3|n$, $5|n$, etc). When one is found, use the dividend and begin again. Repeat.

    Find prime factorization for 7007.
    2, 3, 5 don't divide 7007 but 7 does (1001)
    Now, 7 also divides 1001 (143)
    7 doesn't divide 143 but 11 does (13) and we're done.

Primes

- **Theorem :** If a natural number n >1 is not prime, then n is divisible by some prime number $p \leq \sqrt{n}$.

Given any natural number n > 1, there exists a prime p such that p | n.

*** An integer is prime if it is not divisible by any prime less than or equal to its square root.

Example. What primes must you divide 127 by to test whether it is prime?

To see whether 127 is prime, I only need to see if it has a prime factor

$$\leq \sqrt{127} \approx 11.27$$

You can do the arithmetic to verify that 127 isn't divisible by 2, 3, 5, 7, or 11.

Hence, it must be prime.

Congruence

- Let $n > 1$ be a fixed natural number.
- Given integers a and b, a is **congruent** to be modulo n (or a is congruent to b mod n for short) $a \equiv b \pmod{n}$,
- If and only if $n \mid (a - b)$.
- n is called the modulus of the congruence

Congruence

Then *a* **mod** *b* denotes the remainder *r* from the division “algorithm” with dividend *a* and divisor *b*

- $3 \equiv 17 \pmod 7$ because $3 - 17 = -14$ is divisible by 7;
- $-2 \equiv 13 \pmod 3$, because $-2 - 13 = -15$ is divisible by 3;
- $60 \equiv 10 \pmod{25}$, because $60 - 10 = 50$ is divisible by 25;
- $-4 \equiv -49 \pmod 9$, because $-4 + 49 = 45$ is divisible by 9;

Divisibility

Example: Determine

Whether 17 is congruent to 5 modulo 6, and

Whether 24 and 14 are congruent modulo 6.

Solution:

Divisibility

Theorem: Congruence modulo n satisfies the following:

1. $a \equiv a$ for any a;
2. $a \equiv b$ implies $b \equiv a$;
3. $a \equiv b$ and $b \equiv c$ implies $a \equiv c$;
4. $a \equiv 0$ iff $n \mid a$;
5. $a \equiv b$ and $c \equiv d$ implies $a+c \equiv b+d$;
6. $a \equiv b$ and $c \equiv d$ implies $a-c \equiv b-d$;
7. $a \equiv b$ and $c \equiv d$ implies $ac \equiv bd$;
8. $a \equiv b$ implies $a^j \equiv b^j$ for each integer $j \geq 1$

Real life example:
the clock system

Congruence

FERMAT'S LITTLE THEOREM If p is prime and a is an integer not divisible by p, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Furthermore, for every integer a we have

$$a^p \equiv a \pmod{p}.$$

1. Pick some values a and p which meet the requirements above. Does the theorem hold?

2. Use Fermat's Little Theorem to find 7^{222} mod 11.

Arithmetic modulo m

- Let $Z_m = \{0, 1, \ldots, m-1\}$.
- The operation $+_m$ is defined as $a +_m b = (a + b) \mod m$.
  This is addition modulo m.
- The operation $\cdot_m$ is defined as $a \cdot_m b = (a \cdot b) \mod m$.
  This is multiplication modulo m.
- Using these operations is said to be doing arithmetic modulo m.

Example: Find $7 +_{11} 9$ and $7 \cdot_{11} 9$.

Arithmetic modulo m

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Example: Find $7 +_{11} 9$ and $7 \cdot_{11} 9$.
Solution: Using the definitions above:
$7 +_{11} 9 = (7+9) \mod 11 = 16 \mod 11 = 5$
$7 \cdot_{11} 9 = (7 \cdot 9) \mod 11 = 63 \mod 11 = 8$

Congruence

- Two integers are *relatively prime* if their only common factor is 1, gcd(a,b) = 1

Let $a \in \mathbb{Z}_n$. The multiplicative inverse of a modulo n is an integer $x \in \mathbb{Z}_n$ such that ax ≡ 1 (mod n).

If such an x exists, then it is unique, and a is said to be invertible, or a unit; the inverse of a is denoted by a−1.

Let $a \in \mathbb{Z}_n$. Then a is invertible if and only if gcd(a, n) = 1.

The invertible elements in $\mathbb{Z}_9$ are 1, 2, 4, 5, 7, and 8.

For example,

4−1 = 7 because 4 · 7 ≡ 1 (mod 9).

Congruence

Euler phi function

For $n \geq 1$, let $\varphi(n)$ denote the number of integers in the interval $[1, n]$ which are relatively prime to n.

The function φ is called the Euler phi function (or the Euler totient function).

properties of Euler phi function

(i) If p is a prime, then $\varphi(p) = p - 1$.

(ii) The Euler phi function is multiplicative. That is, if $\gcd(m, n) = 1$, then
$\varphi(mn) = \varphi(m) \cdot \varphi(n) = (m-1)*(n-1)$

Congruence

Example n=15

$Z_{15} = \{1,2,4,6,7,8,11,13,14\}$

$$
\begin{aligned}
|Z_{15}| &= \varphi(15) \\
&= \varphi(3*5) \\
&= \varphi(3)*\varphi(5) \\
&= (3-1)*(5-1) \\
&= 8
\end{aligned}
$$

Euler's theorem : If $a \in Z_n$, then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Thank you

End of Chapter 4